Architecture-Driven Speed–Accuracy Trade-off in Monocular Depth Estimation Under Edge Deployment Constraints

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***Abstract* — Monocular depth estimation offers a low-cost alternative to hardware-based sensing for assistive navigation, but achieving real-time performance on edge-constrained systems remains a critical challenge. This study evaluates whether model architecture, rather than input resolution, governs the trade-off between inference speed and structural consistency under a strictly controlled and reproducible experimental protocol.**

A transformer-based model (Depth Anything) and a CNN-based model (MiDaS Small) were compared across matched resolutions using a locked RTX 2050 benchmark with cross-device validation. The transformer model maintained low structural error (ROI RMSE < 1.0), while the CNN model exhibited severe degradation (ROI RMSE > 100) under identical conditions. In contrast, the CNN achieved higher throughput, reaching 19.42 FPS at 854×480 compared to 6.96 FPS for the transformer, but failed to preserve reliable depth structure.

These results demonstrate that architectural design, rather than resolution scaling, is the dominant factor governing the speed–consistency trade-off. Crucially, no configuration simultaneously satisfies both real-time performance (>15 FPS) and acceptable structural accuracy, highlighting a fundamental limitation in current monocular depth estimation approaches for safety-critical edge deployment.

Index Terms— Monocular Depth Estimation, Vision Transformers, Convolutional Neural Networks, Architecture Comparison, Edge Deployment, Assistive Navigation, Real-Time Inference, Reproducibility.

I. Introduction

Monocular depth estimation infers 3D scene structure from a single RGB image and offers an affordable alternative to hardware-intensive navigation systems such as LiDAR or stereo cameras for visually impaired users [1], [2]. Although deep learning has improved global depth consistency, modern models impose significant computational constraints that limit real-time edge deployment [3].

Inference latency is safety-critical in assistive navigation. Human auditory response and gait termination require reaction time, and system latency exceeding approximately 200 ms reduces the available safety buffer [4], [5].

Vision Transformer (ViT) models such as DPT have demonstrated superior depth accuracy compared to traditional CNN-based approaches but often introduce substantial computational overhead [6], [7]. While existing optimisation strategies largely focus on model compression, controlled comparisons between model architectures under deployment constraints remain limited. This study treats model architecture as the primary independent variable to evaluate its effect on inference throughput and Region of Interest (ROI) depth consistency under deployment constraints. Using a strictly aligned evaluation protocol, a transformer-based model and a CNN-based model were compared at matched resolutions to determine whether architectural design can meaningfully influence the speed–consistency trade-off.

The results show clear differences in behaviour: while the CNN baseline achieves faster inference speeds, it suffers from severe structural degradation at lower resolutions. By contrast, the transformer model maintains critical depth consistency but struggles with real-time performance. These findings indicate that the trade-off between inference speed and accuracy is governed by architectural design rather than input scaling alone.

II. Literature Review

A. From Convolutional Networks to Transformers

Early monocular depth estimation systems predominantly employed Convolutional Neural Networks (CNNs) [3], [6]. Based on local receptive fields and hierarchical feature extraction, these architectures are computationally efficient and suitable for higher frame-rate inference on constrained hardware. However, CNN-based approaches can struggle to

capture long-range spatial dependencies, leading to weaker global scene consistency [7].

Vision Transformer (ViT) architectures substantially shifted this paradigm. Models such as the Dense Prediction Transformer (DPT) use self-attention to model global context across the image [7], achieving lower Root Mean Square Error (RMSE) than CNN baselines in comparative studies [6], [7]. This accuracy improvement, however, introduces increased computational overhead due to attention mechanisms and architectural complexity [3], [7], often resulting in sub-real-time performance on edge-constrained devices [3], [7].

B. Architectural Efficiency and Deployment Constraints

To address deployment constraints, recent work has focused on architectural optimisation and model compression [3]. Knowledge distillation transfers representations from large transformer backbones to lighter student models, while quantisation and pruning reduce parameter precision and computational load [1], [3].

However, despite these optimisation strategies, there remains limited controlled evaluation of how different model architectures behave under identical experimental conditions in deployment-oriented scenarios [3]. In particular, transformer-based models are widely recognised for superior global scene understanding and depth consistency, their practical feasibility on constrained hardware remains unclear. Conversely, CNN-based models are generally more efficient, but this efficiency may be achieved at the expense of structural reliability [3], [7].

C. *Latency and Safety in Assistive Navigation*

In assistive navigation, latency must be treated as a safety constraint rather than a flexible performance metric. Human auditory response time is approximately 0.24 seconds, and gait termination requires additional reaction time [4], [5]. When system latency exceeds ~200 ms, the available safety margin decreases, increasing collision risk [1], [2].

Practical systems should therefore target inference speeds above approximately 15 FPS to maintain responsive guidance under safety constraints [1], [2], [4], [5]. Although the literature frequently reports improvements in depth accuracy, fewer studies evaluate whether such models satisfy strict real-time deployment thresholds on edge hardware, particularly within ground-level safety-critical regions [1], [2].

III. Test Configuration

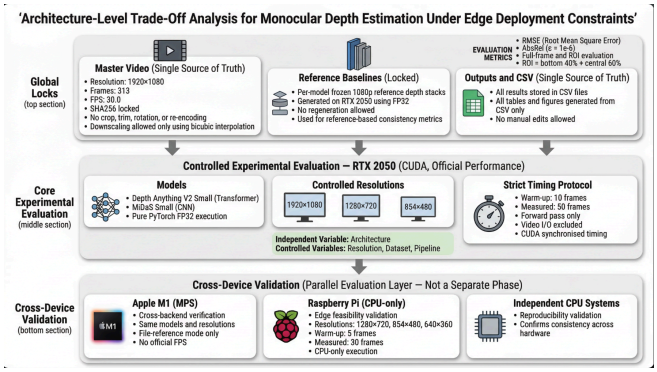

Fig. 1. Locked experimental protocol showing architecture comparison and validation setup.

All experiments followed a strictly locked, reproducible protocol in which model architecture was treated as the primary independent variable while input resolution remained controlled. The master video (S2_Main_Approach_h264.mp4) was SHA256-locked to prevent dataset drift. A 1920×1080 reference depth stack was generated once on an RTX 2050 (CUDA, FP32) and reused in file-reference mode across all experiments.

The Region of Interest (ROI) was defined as the bottom 40% and central 60% of each frame to represent safety-critical ground-level regions. Accuracy was evaluated using ROI RMSE and ROI AbsRel ($\varepsilon = 1e{-}6$). Bicubic interpolation was used for all resizing operations. CSV outputs were treated as the single source of truth and were not manually modified.

A. Core Benchmark (RTX 2050)

The official benchmark was executed on an NVIDIA RTX 2050 (CUDA backend) using two architectures:

- Transformer: Depth Anything
- CNN: MiDaS

Both models were evaluated at identical resolutions:

- 1920×1080 (reference baseline)
- 1280×720
- 854×480

Performance was measured over 50 frames using forward-pass-only execution with CUDA-synchronised timing. Only RTX measurements are used for performance claims and architectural comparison.

B. *Edge Feasibility and Cross-Device Validation*

A validation layer was implemented to assess reproducibility and deployment behaviour across heterogeneous systems.

Validation was conducted on multiple independent systems with heterogeneous hardware configurations, including integrated graphics (Intel Iris Xe), 8 GB RAM, and 11th Gen

Intel i5 processors, ensuring that the evaluation pipeline was tested across diverse execution environments.

Apple M1 (MPS backend) was used for cross-backend verification at matched resolutions (1920×1080, 1280×720, and 854×480), ensuring that metric computation and reference alignment remain consistent outside the CUDA environment.

Raspberry Pi 4 was used for feasibility testing under constrained hardware conditions.

These results are reported strictly as validation and do not redefine the official RTX benchmark.

IV. Results

A. Core benchmark (RTX 2050 — Architecture Comparison)

TABLE I:
RTX 2050 results for transformer and CNN models.

Model	Architecture	Resolution	FPS	RMSE (ROI)	AbsRel (ROI)
Depth Anything	Transformer	1920×1080	0.40	0.00	0.00
Depth Anything	Transformer	1280×720	1.80	0.27	0.057
Depth Anything	Transformer	854×480	6.96	0.96	0.191
MiDaS	CNN	1920×1080	4.09	0.00	0.00
MiDaS	CNN	1280×720	8.73	96.84	0.381
MiDaS	CNN	854×480	19.42	101.02	0.355

Higher FPS indicates better performance, while lower RMSE and AbsRel indicate better accuracy. Across all evaluated resolutions, the transformer model maintains consistently low error, with ROI RMSE remaining below 1.0. In contrast, the MiDaS CNN model exhibits substantially higher error values exceeding 90–100 RMSE, representing approximately two orders of magnitude difference in structural error. This indicates that the observed performance gap is fundamentally driven by architectural design rather than incremental variation.

The results also show a clear architectural divergence in throughput. The CNN model achieves substantially higher FPS across all resolutions. At 854×480, MiDaS reaches 19.42 FPS, compared to 6.96 FPS for the transformer model, representing approximately a 2.8× increase in throughput. However, this performance gain is accompanied by severe structural degradation, with ROI RMSE increasing from 0.96 to 101.02, representing over a 100× increase in error under identical conditions.

These findings confirm that improved computational efficiency does not translate to reliable depth estimation, highlighting a fundamental trade-off between speed and structural consistency.

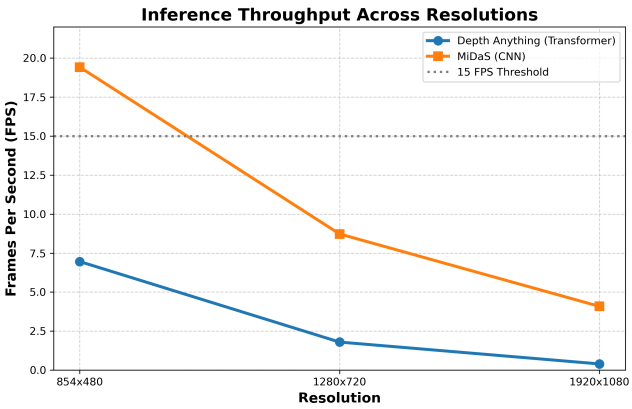

Fig. 2. Inference throughput across resolutions.

The CNN-based model achieves substantially higher throughput across all resolutions, reaching 19.42 FPS at 854×480 compared to 6.96 FPS for the transformer model. This performance gap persists at higher resolutions, with the CNN achieving 4.09 FPS at 1920×1080 while the transformer operates at only 0.40 FPS.

Only the CNN model exceeds the approximate real-time threshold (15 FPS), and only at the lowest resolution (854×480). However, this gain in throughput is accompanied by severe structural degradation, indicating that increased speed does not translate to deployable performance under safety-critical constraints.

This highlights a critical limitation, as no configuration achieves both real-time performance and acceptable depth consistency.

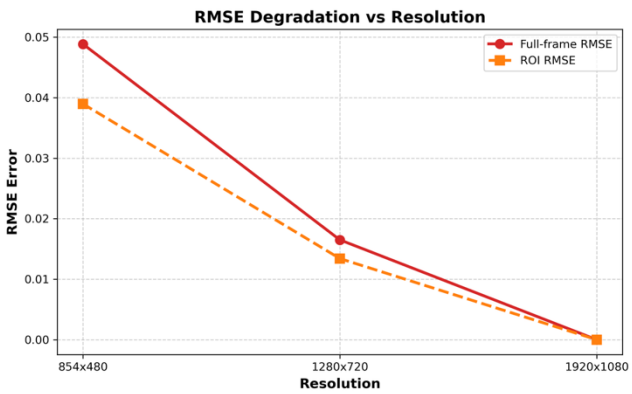

Fig. 3. ROI RMSE across resolutions.

Fig. 3 shows that the transformer-based model maintains low and stable error across all resolutions, with ROI RMSE increasing from 0.00 at 1920×1080 to 0.96 at 854×480, indicating controlled degradation under reduced input resolution.

In contrast, the CNN-based model exhibits substantially higher error, with ROI RMSE values exceeding 96–101 across lower resolutions. This reflects a severe loss of structural consistency within the safety-critical region.

These results highlight a fundamental architectural limitation, where increased computational efficiency in the CNN model comes at the cost of significantly degraded depth accuracy.

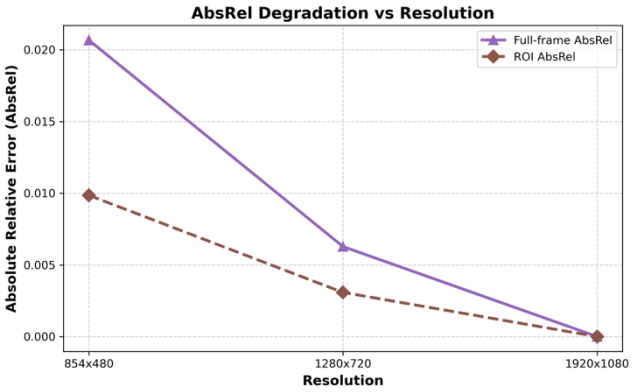

Fig. 4. ROI AbsRel across resolutions.

Fig. 4 shows that AbsRel error increases as resolution decreases, indicating gradual degradation in depth estimation accuracy. The transformer model maintains relatively low error across all resolutions, while the CNN model exhibits significantly higher values, confirming reduced reliability within the safety-critical region.

Fig. 5. Speed–accuracy trade-off across architectures and resolutions (Pareto analysis).

Fig. 5 illustrates the trade-off between inference speed and structural consistency across architectures and resolutions. The CNN model achieves higher throughput, reaching up to 19.42 FPS, but exhibits a substantial increase in error. In contrast, the transformer model maintains low error (RMSE < 1.0) but operates at significantly lower speeds. No configuration satisfies both the real-time threshold (15 FPS) and acceptable accuracy, indicating the absence of a Pareto-optimal solution under the evaluated conditions.

B. Validation and Feasibility Layer

The validation and feasibility layer evaluates model behaviour across non-CUDA backends and heterogeneous systems, and does not replace the core RTX benchmark.

On Raspberry Pi 4 (CPU-only), both architectures exhibit extremely limited throughput (Table II). The transformer model achieves 0.024 FPS at 1280×720, increasing to 0.165 FPS at 640×360, while the CNN model reaches 0.052 FPS at 1280×720 and 0.212 FPS at 640×360. Despite this relative improvement, performance remains well below real-time requirements.

These results remain several orders of magnitude below real-time requirements (≥15 FPS), confirming that CPU-only deployment is fundamentally infeasible under the evaluated configurations and that hardware constraints alone cannot compensate for architectural inefficiency.

In terms of structural consistency, the transformer model maintains moderate degradation, with ROI RMSE increasing from 0.35 (720p) to 0.58 (640×360). In contrast, the CNN model exhibits significantly higher error, with ROI RMSE exceeding 107 at 720p and increasing to 135.92 at 640×360, indicating severe instability under constrained execution.

Apple M1 (MPS backend) demonstrates improved performance relative to CPU-only systems but remains below real-time thresholds. Importantly, ROI consistency trends closely match those observed in the RTX benchmark, confirming that metric computation and reference alignment remain stable across different hardware backends.

Additional validation was conducted on independent CPU-based systems with integrated graphics. These systems reproduced consistent trends, with the transformer model maintaining low error and the CNN model exhibiting substantially higher error across all resolutions. This cross-device agreement confirms that the observed architectural behaviour is reproducible and not dependent on a specific hardware configuration.

These results further reinforce that the observed performance limitations are inherent to model architecture rather than hardware-specific artefacts.

TABLE II
Raspberry Pi 4 CPU feasibility results for transformer and CNN models.

Model	Resolution	FPS	RMSE (ROI)	AbsRel (ROI)
Depth Anything	1280×720	0.024	0.35	0.060
Depth Anything	640×360	0.165	0.58	0.116
MiDaS	1280×720	0.052	107.97	0.396
MiDaS	640×360	0.212	135.92	0.548

Table II shows that both architectures perform far below real-time requirements on CPU-only hardware, with the transformer maintaining lower error while the CNN exhibits severe degradation.

C. Discussion

1) Cross-Device Reproducibility

Validation across heterogeneous systems, including Apple M1 (MPS), Raspberry Pi (CPU), and additional independent CPU-based environments, produced consistent metric behaviour relative to the RTX baseline. For example, the transformer model maintained ROI RMSE below 1.0 on RTX and within 0.35–0.58 on Raspberry Pi, while the CNN model consistently exceeded 90–135 RMSE across all devices.

This cross-device agreement confirms that the evaluation pipeline, reference-based comparison strategy, and ROI definitions are robust and hardware-agnostic. The observed trends can therefore be attributed to architectural behaviour rather than backend-specific effects.

2) Architectural Trade-off Behaviour

The results clearly demonstrate that model architecture is the dominant factor governing the trade-off between inference speed and structural consistency. The transformer-based model maintains low error across all resolutions, with ROI RMSE remaining below 1.0, while the CNN model exhibits significantly higher error exceeding 100 RMSE under identical conditions.

In contrast, the CNN model achieves substantially higher throughput, reaching 19.42 FPS at 854×480, compared to 6.96 FPS for the transformer model. This represents a 2.8× increase in speed, but at the cost of an over 100× increase in error, highlighting a fundamental architectural trade-off between efficiency and reliability.

Importantly, the CNN model does not exhibit gradual degradation, but instead demonstrates structural instability under resolution reduction, indicating a failure to preserve depth relationships rather than a controlled loss of precision.

3) Limited Impact of Resolution under Architectural Constraints

In contrast to the preceding phase of this research, where input resolution was examined as the sole independent variable, the present findings indicate that resolution plays a secondary role once architectural design is introduced as a controlled factor.

The results indicate that input resolution has a secondary impact compared to architectural design. While the transformer model shows controlled degradation from 0.27 RMSE at 1280×720 to 0.96 RMSE at 854×480, this change remains limited relative to the extreme error observed in the CNN model, which consistently exceeds 96–101 RMSE across all evaluated resolutions.

Similarly, although throughput increases with downsampling, the gains are insufficient to meet real-time constraints. For example, the transformer model improves from 1.80 FPS at 1280×720 to 6.96 FPS at 854×480, yet still remains below the ~15 FPS threshold required for safety-critical deployment. This demonstrates that resolution scaling cannot compensate for inherent architectural overhead, particularly in transformer-based dense prediction models.

4) Safety-Critical Implications for Assistive Navigation

From a deployment perspective, neither architecture provides a configuration that satisfies the combined requirements of real-time performance and acceptable structural reliability for responsive assistive navigation systems. The transformer model remains significantly below this threshold, achieving a maximum of 6.96 FPS, while the CNN model reaches up to 19.42 FPS at lower resolutions but introduces severe structural degradation, with ROI RMSE exceeding 100.

This trade-off presents a critical limitation: models capable of maintaining accurate depth estimation operate at impractically low speeds, while models achieving higher throughput fail to provide reliable structural information within safety-critical regions. As a result, neither configuration is suitable for real-time deployment in assistive navigation scenarios under the evaluated conditions.

5) Empirical Confirmation of Theoretical Limitations

The findings provide empirical support for existing literature that identifies a tension between accuracy and efficiency in modern depth estimation architectures. Transformer-based models deliver superior consistency but incur significant computational cost [7], while CNN-based models offer higher computational efficiency, but under the evaluated conditions this efficiency is associated with substantially reduced structural reliability.

This study extends prior work by quantifying this trade-off under a strictly controlled and reproducible framework, demonstrating that the limitation persists even when external variables are tightly constrained.

6) Key Insight and Research Direction

The central insight from this study is that architectural design, rather than input resolution, determines the achievable balance between inference speed and structural consistency. Across all evaluated configurations, no model achieved both acceptable accuracy (RMSE < 1.0) and real-time performance (>15 FPS), confirming that current approaches are fundamentally constrained by architectural limitations.

The absence of a Pareto-optimal configuration highlights a critical gap between model capability and practical

deployment requirements. These findings indicate that improvements in input-level optimisation, such as resolution scaling, are insufficient to overcome inherent architectural overhead.

This motivates a shift in research focus towards architectural efficiency as the primary optimisation objective. Future work will preserve the locked dataset, ROI definition, and evaluation protocol while exploring lightweight transformer variants, hybrid CNN–transformer architectures, and efficiency-driven model adaptations. The objective is to identify architectures capable of achieving a meaningful shift in the speed–consistency trade-off towards real-time, safety-critical deployment.

V. Conclusion/Future Work

This study presents a controlled evaluation of architectural efficiency in monocular depth estimation under a strictly locked and reproducible experimental protocol. By isolating model architecture as the primary independent variable, the results demonstrate that architectural design fundamentally governs the trade-off between inference speed and structural consistency.

The transformer-based model consistently preserved depth accuracy within the safety-critical region of interest, confirming its strong capability for global scene representation. In contrast, the CNN-based model achieved higher throughput but exhibited substantially higher error, indicating reduced structural reliability. Despite these differences, neither architecture provided a configuration that simultaneously satisfied real-time performance requirements and acceptable structural reliability. This reveals a persistent gap between model capability and practical deployability. Validation across heterogeneous systems, including CPU-based environments and Apple M1, confirmed that the observed trends are reproducible and not hardware-specific. This consistency strengthens the reliability of the experimental framework and supports the generality of the findings.

From a deployment perspective, the results indicate that neither input resolution scaling nor existing architectural configurations alone are sufficient to meet the constraints of real-time, safety-critical assistive navigation. This reinforces the need to prioritise architectural efficiency as the central optimisation objective.

Future work will preserve the locked dataset, ROI definition, and evaluation protocol while introducing targeted architectural optimisation strategies, including lightweight transformer variants and hybrid CNN–transformer designs. The objective is to identify architectures capable of achieving a meaningful shift in the speed–consistency trade-off towards practical edge deployment.

Ultimately, this work establishes that architectural efficiency remains the principal bottleneck preventing the real-world deployment of monocular depth estimation systems in safety-critical environments.

VI. References

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